

Before the Levy: Announcement Pass-Through and Carbon-Price Calibration in Denmark’s Livestock Tax

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Abstract

Denmark announced in June 2024 that it would introduce a greenhouse-gas tax on livestock emissions. I estimate the anticipatory response of beef prices using two independent sources: scraped product–store prices and the official Danish consumer price index (CPI). A product–store difference-in-differences (DiD) design yields an average beef-price increase of 4.4 log points. A lags-only DiD applied to official CPI data yields 6.4 log points. The estimates differ by only 0.6 standard errors under an approximate cross-source test. Aggregate synthetic DiD is positive but imprecise. All specifications end in September 2025, before the official October 2025 onset of a bovine viral diarrhoea outbreak; the aggregate specifications use symmetric 15-month pre- and post-periods. I then pool ten comparable social-cost-of-carbon (SCC) estimates, treating otherwise undated estimates as time-invariant and propagating published model ranges. The equal-paper-weight mean is USD 158/tCO₂ in 2024 dollars, with a hierarchical 95 percent interval of USD 77–363. The estimated announcement response is economically meaningful but smaller than the price adjustment implied by that pooled benchmark. The agreement across independent DiD estimates supports the use of scraped prices for timely policy evaluation, while the estimator dispersion and SCC uncertainty counsel against interpreting any single calibration as definitive.

1 Introduction

Pricing agricultural greenhouse-gas emissions is difficult because the emissions are biologically heterogeneous, farm-level measurement is costly, and food prices are politically salient. A further complication is timing. If a future tax is credible, firms may revise contracts, procurement, and retail prices when the policy is announced rather than when the first liability is paid. Estimates that begin only at implementation can therefore miss part of the policy response.

On 24 June 2024, the Danish government and the parties to the Green Tripartite announced an agreement to introduce a greenhouse-gas tax on livestock emissions (Danish Government, 2024). The measure is an upstream agricultural emissions tax rather than a retail meat excise, and implementation was scheduled for later years. This paper asks whether the announcement changed supermarket beef prices and how the estimated response compares with the price signal implied by the broader SCC literature.

The empirical contribution is to triangulate the same research question with two sources whose weaknesses are complementary. The scraped data provide a large, timely product–store panel but are non-official and require transparent text classification and unit normalization. Statistics Denmark’s CPI is official and consistently classified, but supplies only a small cross-section of product indices. The microdata DiD estimate is 0.0437 (standard error 0.0163); the aggregate lags-only DiD estimate is 0.0645 (HAC standard error 0.0306). Their difference is not statistically meaningful. Synthetic DiD provides a complementary aggregate robustness check and gives 0.0760, although its placebo interval is wide.

The second contribution is a literature-based calibration that does not privilege a single SCC study. I harmonize one preferred global total-SCC estimate per paper to 2024 dollars. Estimates reported for years other than 2030 enter when the paper does not imply a time path, in which case the estimate is treated as time-invariant. Published uncertainty ranges are retained and propagated through a hierarchical paper bootstrap. This produces a ten-paper mean of USD 158/tCO₂ and a wide interval of USD 77–363. The exercise is a transparent synthesis of heterogeneous studies, not a claim that their model ranges are conventional sampling confidence intervals.

The broader contribution concerns announcements as policy instruments. A credible commitment can alter expectations and behaviour before the underlying instrument is deployed, an idea made especially visible by Mario Draghi’s *whatever it takes* commitment during the euro-area crisis (Draghi, 2012). The analogy is not exact: this paper studies prices in a commodity market, not central-bank credibility. It nevertheless motivates separating announcement effects from implementation effects and asking whether an announced future carbon price generates measurable adjustment in advance.

The paper focuses exclusively on beef. The estimand is especially relevant because cattle are the most emissions-intensive major livestock product and prominent in the Danish agreement. Restricting attention to a pre-specified commodity also avoids selecting among a large family of weakly identified commodity-specific estimates. The remainder of the paper describes the two data sources and classification rules, sets out the three estimators, reports cross-source results, and uses the SCC synthesis for a back-of-the-envelope policy calibration.

2 Data and Panel Construction

2.1 Scraped grocery-price microdata

The first source is the Danish grocery-price history assembled by dagligepriser.dk. A raw record identifies a store, product, product description, date, price, and—when available—package quantity and unit. I retain observations with a valid positive price and date. Repeated observations within a product–store–month are averaged, so the unit of observation for the main panel is a product–store pair in a calendar month.

Prices are converted to DKK per kilogram for mass units and DKK per litre for volume units. Reported package quantities take precedence. When quantity or unit fields are absent, the pro-

cedure searches the product description for a number followed by kg, g, litre, ml, or cl. Grams, millilitres, and centilitres are converted to their base units before dividing price by quantity. Items measured only per piece and observations with missing, non-positive, or unsupported quantities are excluded. Because normalized prices remain right-skewed, estimation uses their natural logarithm and Table 1 reports medians and quartiles as well as means.

Product classification is deterministic and auditable. Names and source categories are case-folded; Danish characters and common encoding variants are transliterated; and short tokens are matched at word boundaries. Beef is assigned when the normalized text contains an English or Danish beef term, including *beef*, *okse*, *oksekød*, *kvæg*, *kalv*, *veal*, *entrecôte*, or *roastbeef*. The comparison group consists of classified foods not exposed as livestock treatment items: poultry; fish and seafood; eggs; fruit and vegetables; grains and bread; plant oils; sweets and snacks; beverages; plant proteins; and pantry goods. Pork, lamb, sheep, goat, dairy, and products matching more than one livestock category are never controls for beef. Prepared foods containing terms such as pizza, lasagne, ready meal, or sandwich are flagged as ambiguous when a livestock term is present. Non-food keywords are removed before commodity matching, and unmatched products are excluded rather than silently assigned to the control group.

The event month, June 2024, is excluded from both pre and post. The scraped panel begins in November 2023 and the analysis retains post-announcement months through September 2025, for 23 observed months in total. October 2025, the sixteenth full post-announcement month, and all later months are excluded. The Danish Veterinary and Food Administration records a bovine viral diarrhoea (BVD) outbreak from October 2025 to May 2026 in three cattle holdings and notes that BVD can reduce appetite and milk output (Danish Veterinary and Food Administration, 2026). A veterinary report states that surveillance detected the affected large dairy herd in September 2025 (Dansk Veterinærtidsskrift, 2025). Ending in September does not prove that all anticipatory disease effects are absent, but it excludes the officially dated outbreak interval and avoids mechanically attributing the subsequent cattle-specific supply disturbance to the tax announcement.

2.2 Official aggregate consumer prices

The second source is Statistics Denmark’s monthly PRIS01 consumer price index by COICOP 2018 product class. Beef and veal (011221) is the treated series. Candidate donors are terminal food and non-alcoholic-beverage classes with complete coverage in the estimation window. I exclude pork; lamb, sheep, and goat; dried or salted meat; offal; meat preparations; milk, yoghurt, cheese, cream, dairy desserts, other dairy products; and butter and other dairy fats. These categories are either directly livestock-exposed or sufficiently close to treated animal products that they could absorb the same announcement response. The resulting balanced panel contains beef and 52 donors.

The official-data window runs from April 2023 through September 2025: 15 months before July 2024 and 15 months from July 2024 onward. June 2024 is the final pre-treatment reference month for event-study purposes. This symmetric window prevents the aggregate estimates from being driven by unequal pre- and post-support and, like the microdata restriction, ends before the official

October 2025 BVD outbreak interval.

Table 1 summarizes both estimation samples. The scraped panel contains 151,933 monthly observations from 16,912 product–store pairs, of which 5,123 observations and 592 pairs are classified as beef. Normalized prices are strongly right-skewed: the overall mean is DKK 518 per kilogram or litre, compared with a median of DKK 100, and the beef mean is DKK 266 compared with a median of DKK 127. This dispersion motivates logarithmic estimation while retaining prices in levels for transparent description. Product availability is intermittent, with the median product–store observed for eight months overall and 7.5 months for beef. The official panel is balanced by construction: 30 observations for beef and 1,560 for the 52 donor series. Its CPI distribution is much tighter, reflecting aggregation within official product classes rather than product-level price heterogeneity.

Table 1: Descriptive statistics for the estimation samples

Sample	Variable	Observations	Mean	SD	P25	Median	P75
<i>Panel A. Scraped product–store prices</i>							
All	Normalized price	151,933	518.19	4,337.95	51.84	100.00	183.33
Beef	Normalized price	5,123	265.64	1,925.50	70.00	127.00	209.75
Controls	Normalized price	146,810	527.00	4,398.05	51.18	100.00	182.14
All	Months per product–store	16,912	8.98	6.26	3	8	14
Beef	Months per product–store	592	8.65	5.79	4	7.5	13
Controls	Months per product–store	16,320	9.00	6.27	3	8	15
<i>Panel B. Statistics Denmark consumer price indices</i>							
All	Consumer price index	1,590	97.76	6.87	95.25	98.73	101.22
Beef	Consumer price index	30	89.43	7.41	84.74	86.58	89.73
Donors	Consumer price index	1,560	97.92	6.77	95.47	98.79	101.24

Source: dagligepriser.dk (Panel A); Statistics Denmark, PRIS01 (Panel B).

Notes: Scraped prices are normalized to DKK per kilogram or litre. For the months-per-product–store rows, observations count product–store units. Panel B covers the balanced April 2023–September 2025 window; its observation count is series–months. Donors exclude livestock-exposed food categories.

3 Empirical Strategy

The microdata DiD is

$$\log p_{it} = \alpha_i + \lambda_t + \beta(\text{Beef}_i \times \text{Post}_t) + \varepsilon_{it}, \quad (1)$$

where p_{it} is the normalized price of product–store pair i in month t ; α_i and λ_t are product–store and month fixed effects; Beef_i identifies beef products; Post_t identifies full months after the announcement; β is the average treatment effect on the treated (ATT); and ε_{it} is the error term. Standard errors are clustered by product–store. The event study replaces the post indicator with relative-month interactions and omits May 2024, the final observed pre-announcement month.

The official-data lags-only DiD preserves the earlier specification. For each month, I subtract the unweighted mean log CPI of the 52 donors from log beef CPI and regress this gap on the post indicator. Newey–West standard errors use two lags. The corresponding official event study

interacts beef with month indicators, controls for calendar month and three- and four-digit COICOP categories, and uses heteroskedasticity-robust standard errors.

Synthetic DiD follows Arkhangelsky et al. (2021), jointly weighting donor series and pre-treatment periods to construct a comparison path for beef. It operates on the balanced official panel and uses placebo inference with 200 replications. All DiD and SDiD estimations and their figures are produced in Stata; Python is limited to downloading, classifying, normalizing, panel construction, and orchestration.

4 Results

Table 2 reports the three beef estimates. The scraped microdata DiD implies a 4.37-log-point increase (approximately 4.47 percent), with a clustered standard error of 1.63 percentage points. The official lags-only DiD gives 6.45 log points, with a Newey–West standard error of 3.06 points. An approximate equality test, treating the estimates as independent, gives

$$z = \frac{0.0645 - 0.0437}{\sqrt{0.0306^2 + 0.0163^2}} = 0.60 \quad (p \approx 0.55).$$

Thus the official and scraped DiD estimates are statistically compatible. This agreement is encouraging for the external validity of the non-official price panel, although it cannot validate every scraped product or its level normalization.

Table 2: Estimated effect of the livestock-tax announcement on beef prices

	(1)	(2)	(3)
ATT	0.0437*** (0.0163)	0.0645** {0.0306}	0.0760 [0.115]
Observations	151,933	1,590	1,590
Units	16,912	53	53
Periods	23	30	30
Pre-treatment average	158.54	84.88	84.88
R^2	0.961	—	—
Time window	2023m11–2025m9	2023m4–2025m9	2023m4–2025m9
Lags only	No	Yes	No

Source: dagligepriser.dk (column 1); Statistics Denmark, PRIS01 (columns 2–3).

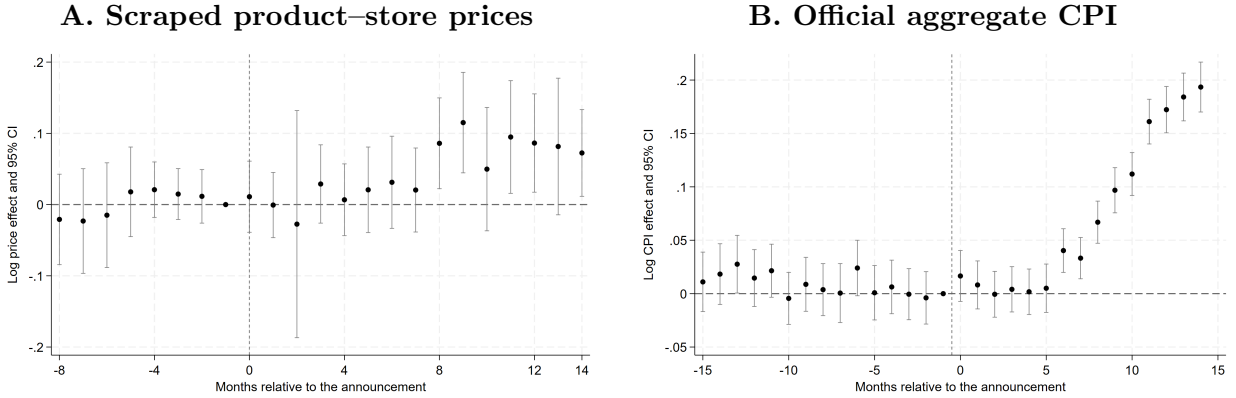
Notes: The outcome in column 1 is log normalized product price; its DiD includes product–store and month fixed effects. Parentheses contain standard errors clustered by product–store. Column 2 applies the unchanged lags-only aggregate DiD to the log CPI gap between beef and the donor mean; braces contain Newey–West HAC standard errors with two lags. Its reported observation count is the 1,590 series–months in the official panel before cross-sectional averaging into 30 monthly gaps. Column 3 is synthetic DiD; brackets contain its placebo p-value. The pre-treatment average is DKK/kg in column 1 and the CPI index in columns 2–3. Other livestock-exposed products are excluded from every beef control group. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Aggregate SDiD is positive at 0.076 but its 95 percent interval, $[-0.018, 0.170]$, includes zero. The three columns therefore support a narrower conclusion than a single preferred estimate would:

the two DiD designs point to a modest positive response, while synthetic reweighting is too imprecise in this small aggregate panel to determine the magnitude precisely.

Figure 1 compares event-time patterns. In the scraped panel, the shorter pre-period is noisier but the pre-announcement coefficients are jointly insignificant ($p = 0.78$), and the post coefficients are generally positive. In the official panel, coefficients are near zero throughout the 15-month pre-period and then rise gradually, especially from the sixth post month. The difference in smoothness is expected: official CPI aggregates many price quotes, whereas the scraped panel contains entry, exit, heterogeneous packaging, and intermittent product availability.

Figure 1: Beef event-study estimates across data sources



Source: dagligepriser.dk (Panel A); Statistics Denmark, PRIS01 (Panel B).

Notes: Points are relative-month coefficients and bars are 95 percent confidence intervals. In both panels, July 2024 is $t = 0$, the first complete post-announcement month. Panel A omits May 2024 ($t = -1$), because the partial event month of June is excluded, and uses product-store clustered standard errors. Panel B uses June 2024 as its omitted reference month and heteroskedasticity-robust standard errors. Both samples end in September 2025, before the official October 2025 BVD outbreak interval.

5 SCC Synthesis and Policy Calibration

The calibration uses one preferred total global SCC from each eligible paper (Cai et al., 2016; Nordhaus, 2017; Ricke et al., 2018; Cai and Lontzek, 2019; Pindyck, 2019; Hänsel et al., 2020; van den Bremer and van der Ploeg, 2021; Dietz et al., 2021; Rennert et al., 2022; Barrage and Nordhaus, 2024). Dollar values are inflated to 2024 US dollars and values per ton of carbon are converted to tons of CO₂. The 2030 value from Barrage and Nordhaus (2024) is log-interpolated between their published 2025 and 2050 path points. For the other eligible studies, a reported value at another date is treated as constant when the source does not provide a time path.

The equal-paper-weight point mean across ten studies is USD 157.96/tCO₂. The interval is constructed in two explicit layers. Index papers by j . Let θ_j be paper j 's preferred harmonized SCC estimate and $[L_j, U_j]$ its reported positive central range. Let α_j be the probability in each tail outside that range, so the range has coverage $1 - 2\alpha_j$. Let Φ denote the standard normal cumulative

distribution function and define

$$z_j = \Phi^{-1}(1 - \alpha_j), \quad \sigma_j = \frac{\log U_j - \log L_j}{2z_j}.$$

where z_j is the corresponding standard-normal quantile and σ_j is the calibrated log-scale standard deviation. For a selected paper with a usable range, the within-paper draw is

$$X_j = \exp(\log \theta_j - \frac{1}{2}\sigma_j^2 + \sigma_j Z_j), \quad Z_j \sim N(0, 1),$$

where Z_j is an independent standard-normal variate, X_j is the simulated SCC value, and $E[\cdot]$ denotes expectation. The $-\sigma_j^2/2$ adjustment ensures that $E[X_j] = \theta_j$, while the range's log width determines dispersion. For papers without a positive range, $X_j = \theta_j$. The 66 percent model range in Ricke et al. (2018) has $\alpha_j = 0.17$ and therefore $z_j = \Phi^{-1}(0.83)$; the 5th–95th quantiles in Rennert et al. (2022) have $\alpha_j = 0.05$ and $z_j = \Phi^{-1}(0.95)$; and the ethical-parameter range in Hänsel et al. (2020) is conservatively assigned 95 percent coverage, so $\alpha_j = 0.025$ and $z_j = \Phi^{-1}(0.975)$.

In the second layer, each of 10,000 bootstrap replications samples ten papers with replacement from the ten-paper pool, makes an independent within-paper draw for every sampled paper, and records the arithmetic mean of those ten draws. Resampling papers captures between-paper dispersion; the lognormal step propagates the reported within-paper range. The 2.5th and 97.5th percentiles of the 10,000 replicated means are USD 77.22 and USD 363.40/tCO₂. Because the inputs mix model, parameter, and distributional uncertainty rather than comparable sampling errors, this interval is a hierarchical sensitivity envelope, not a frequentist confidence interval for a common structural parameter.

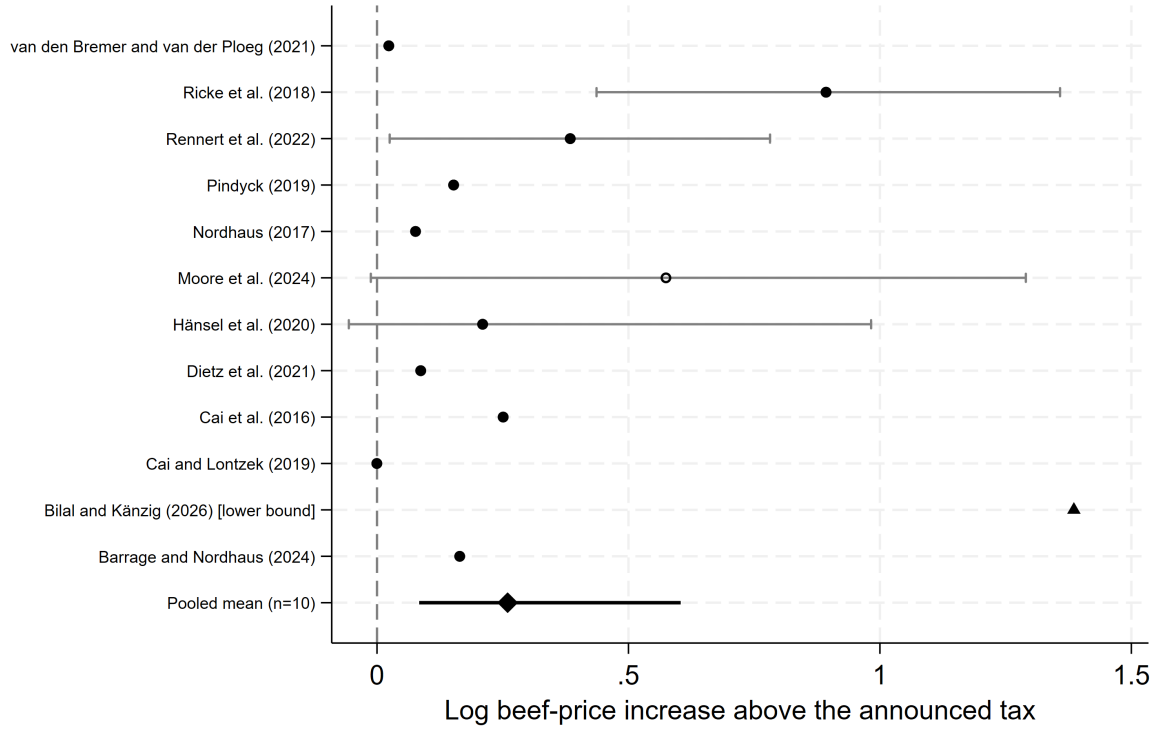
Figure 2 shows each preferred estimate after mapping it into the additional log beef-price increase required above the announced DKK 300/tCO₂ tax. Filled points enter the pool. Hollow points are external comparisons: a partial mortality SCC and a synthesis that could double count the underlying literature (Carleton et al., 2022; Moore et al., 2024). The triangle is the lower bound reported by Bilal and Känzig (2026) and also remains external. The wide intervals reveal why retaining study-level variation materially changes the calibration.

Let $\bar{p} = 158.54$ be the mean pre-announcement beef price in DKK/kg, $e = 6.8953$ DKK/USD, and $q = 59.6$ kg CO₂e/kg beef. The carbon intensity follows the OECD's presentation of the beef-herd benchmark from Poore and Nemecek (2018) (OECD, 2025). For SCC value s in USD/tCO₂ and announced tax $T = 300$ DKK/tCO₂, define $g(s)$ as the additional log price required:

$$g(s) = \log \left[\frac{\bar{p} + (se - T)q/1000}{\bar{p}} \right]. \quad (2)$$

At the pooled mean, $g(157.96) = 0.260$, with a pooled interval of 0.084–0.604. Figure 3 compares that benchmark with all three econometric estimates. The micro DiD confidence interval ends at 0.076, just below the pooled interval's lower endpoint. The aggregate DiD and SDiD intervals overlap its lower tail.

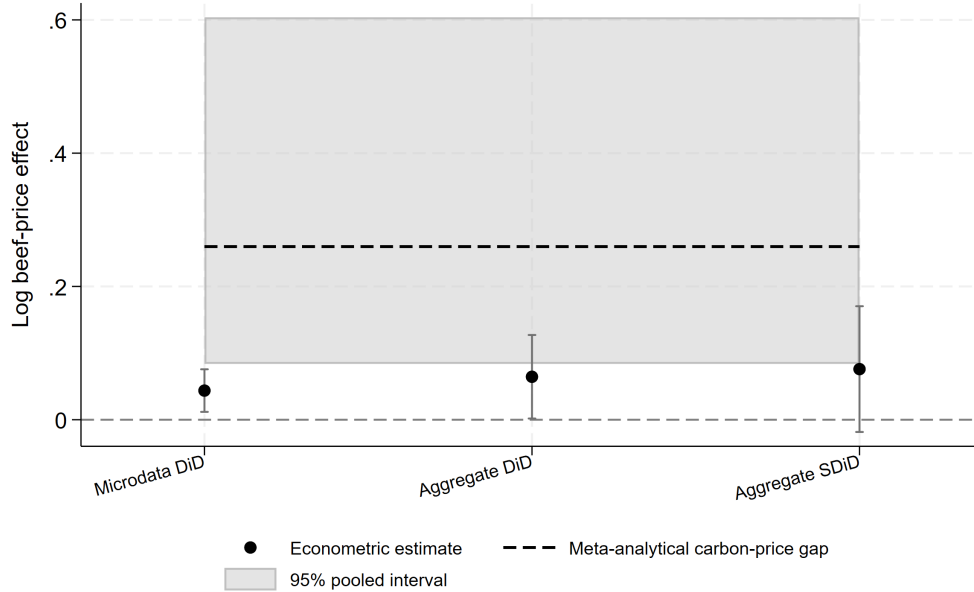
Figure 2: SCC estimates and implied beef-price gaps



Source: The studies listed in Appendix Table A1; OECD (2025).

Notes: Each point maps a preferred SCC estimate into the log beef-price increase needed in addition to the announced Danish tax, using 59.6 kg CO₂e per kg of beef, 6.8953 DKK/USD, and the scraped pre-announcement beef price. Source ranges preserve their original meaning. Filled circles enter the equal-paper-weight pool; hollow circles and the lower-bound triangle do not. The pooled diamond and line show the hierarchical mean and its 95 percent interval.

Figure 3: Econometric estimates and the meta-analytical carbon-price gap



Source: Statistics Denmark, PRIS01; dagligepriser.dk; the studies listed in Appendix Table A1.

Notes: Dots are the three ATT estimates in Table 2; error bars are 95 percent intervals. The dashed line is the log beef-price gap implied by the pooled SCC mean, and the grey band maps the 2.5th and 97.5th percentiles of the 10,000 hierarchical bootstrap means into price space.

The back-of-the-envelope magnitudes are informative but not a welfare calculation. The micro DiD point estimate corresponds to about DKK 7.1/kg at the pre-period mean, or USD 17/tCO₂. Combining this announcement-equivalent signal with the announced gross tax gives roughly USD 61/tCO₂, well below the USD 158 pooled mean. Matching the mean mechanically would require a remaining statutory component of about USD 141/tCO₂, or DKK 970/tCO₂, before deductions and incidence adjustments. This calculation assumes a common beef carbon intensity and full translation of a product-price change into a carbon-price equivalent; neither assumption is structural.

6 Discussion of the Results

Two independent DiD designs find a modest increase in beef prices following Denmark's announcement. Their statistical compatibility is notable because the sources differ sharply in provenance, sampling, and aggregation. It suggests that carefully classified scraped prices can recover economically relevant aggregate movements faster and at much greater cross-sectional scale than official indices alone. That is evidence in favor of reliability and external validity for this application, not a blanket endorsement of non-official data: product coverage is selected, price normalization is noisy in levels, and product availability is intermittent.

Aggregate SDiD prevents an overly tidy conclusion: it is positive but imprecise. With only 53

official series and 15 pre-treatment months, synthetic weighting is intrinsically fragile. The preferred interpretation therefore rests on triangulation: the two DiD estimates agree, their event studies do not show differential pre-trends, and SDiD points in the same direction without determining the magnitude precisely.

The meta-analytical calibration places the estimated announcement effect below the central carbon-price gap while preserving substantial uncertainty across the underlying SCC literature. The lower end of the pooled range overlaps some aggregate econometric intervals, so the evidence does not identify a single corrective tax with useful precision.

The timing of the response is itself policy-relevant. The tax had not yet been implemented, so the estimates are consistent with firms and consumers adjusting to a credible future policy rather than to a current statutory liability. In this respect, the mechanism resembles the expectations channel highlighted by Draghi’s *whatever it takes* commitment: the announcement can influence behaviour before the full instrument is used (Draghi, 2012). The analogy should not be pushed too far, and the design cannot isolate every expectations mechanism. Still, it suggests that clear, credible policy announcements can be instruments in their own right. They need not replace the eventual tax; instead, their measurable anticipatory effects should enter subsequent policy calibration.

Three limitations are central. First, the announcement was national and there is no untreated Denmark; identification is relative to other food prices and depends on the absence of beef-specific shocks before October 2025. Second, text rules and quantity parsing can misclassify products or create extreme normalized levels, although logs, fixed effects, explicit exclusion rules, and agreement with official CPI mitigate this concern. Third, SCC papers estimate heterogeneous objects under different discounting, damage, and uncertainty assumptions. The meta-analysis represents that heterogeneity rather than resolving it.

7 Conclusion

Denmark’s livestock-tax announcement was followed by a modest increase in beef prices in two independent data sources. The close correspondence between the scraped and official DiD estimates supports the use of carefully classified web-scraped prices for timely policy evaluation, while the wider SDiD interval cautions against a precise causal magnitude. Most importantly, the evidence indicates that credible policy announcements may themselves induce behaviour before implementation. Announcement and implementation are therefore distinct policy stages, and evaluations that begin only when a tax is collected may miss an economically meaningful part of the response.

The pooled SCC benchmark leaves a sizeable but uncertain carbon-price gap; its lower tail overlaps the aggregate estimates. Post-implementation work should revisit the calibration using realized incidence, farm-level emissions, and longer panels that separate announcement effects from later supply shocks.

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A Additional Results

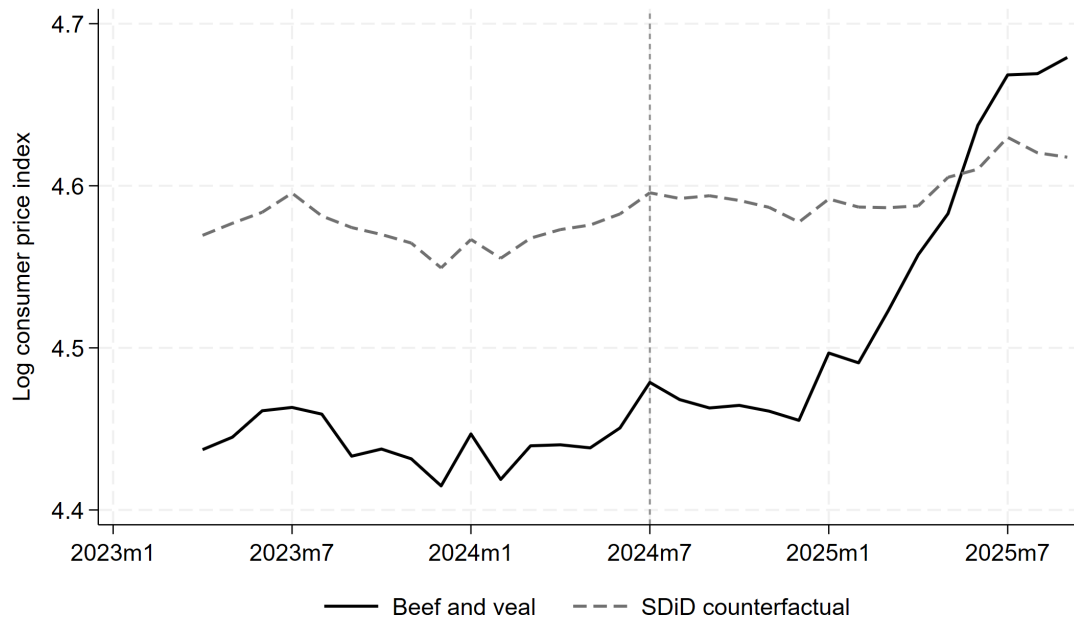
Table A1: Preferred SCC benchmarks and pooling status

Study	Method	Reference year	2024 USD/tCO ₂	Low	High	Pool status
Cai et al. (2016)	Stochastic IAM	2015	153.5	—	—	Included
Nordhaus (2017)	DICE IAM	2030	74.2	—	—	Included
Ricke et al. (2018)	Spatial IAM	2020	599.9	254.6	1,158.0	Included
Cai–Lontzek (2019)	Stochastic IAM	Undated	43.4	—	—	Included
Pindyck (2019)	Expert elicitation	Undated	107.0	—	—	Included
Hänsel et al. (2020)	Updated DICE	2020	133.7	22.5	688.2	Included
van den Bremer–van der Ploeg (2021)	Asset-pricing IAM	2015	52.7	—	—	Included
Dietz et al. (2021)	Meta-analytic IAM	2020	78.5	—	—	Included
Rennert et al. (2022)	Modular IAM	2020	224.2	53.3	500.6	Included
Barrage–Nordhaus (2024)	DICE-2023	2030	112.5	—	—	Included
Carleton et al. (2022)	Mortality damages	2020	21.0	−30.3	65.8	Partial; external
Moore et al. (2024)	Literature synthesis	2020	343.0	38.8	1,059.3	Double-count risk; external
Bilal–Känzig (2026)	Macro empirical	2024	> 1,200	—	—	Lower bound; external

Source: The studies listed in the first column.

Notes: Values are converted to 2024 US dollars per ton of CO₂. One preferred global total-SCC estimate enters per eligible paper. Values at dates other than 2030 are treated as time-invariant when no within-paper path is reported. Barrage–Nordhaus is log-interpolated to 2030. Low and high retain their source-specific interpretation. The pooled mean gives equal weight to each included paper; partial estimates, a literature synthesis, and a lower bound are external comparisons.

Figure A1: Official-data synthetic DiD: beef and SDiD counterfactual



Source: Statistics Denmark, PRIS01.

Notes: Series are log CPI for beef and veal and the Stata SDiD counterfactual over the symmetric 15-month pre/post window. The dashed vertical line marks July 2024, the first post-announcement month.